A logo for Narayanpal

Subhodeep Chakraborty[*], Shiven Bajpai[†] and G.V.V. Sharma[‡]

Question: Given that

$$f(t) = e^{-at}u(t) + e^{bt}u(-t) \tag{0.1a}$$

$$u(t) = \begin{cases} 0, & t < 0 \\ \frac{1}{2}, & t = 0 \\ 1, & t > 0 \end{cases} \tag{0.1b}$$

$$\int_{-\infty}^{\infty} f(t) = 1 \tag{0.1c}$$

Find the possible values of (a,b) if these are the end points of the latus recta of the associated conic. Plot $f(t)$ for these values of (a,b)

Solution: We expand the integral as

$$\int_{-\infty}^{\infty} f(t) = \int_{-\infty}^{0} f(t) + \int_{0}^{\infty} f(t) \tag{0.2}$$

$$= \int_{-\infty}^{0} e^{bt} + \int_{0}^{\infty} e^{-at} \tag{0.3}$$

$$= \frac{1}{b} + \frac{1}{a} \tag{0.4}$$

Subsituting (0.1c) in (0.4)

$$\frac{1}{a} + \frac{1}{b} = 1 \tag{0.5}$$

$$ab - a - b = 0 \tag{0.6}$$

Which is the equation of a conic. If we take a to be x and b to be y and then express this as a conic in standard form (A.15) we get

$$\mathrm{g}(\mathbf{x}) = \mathbf{x}^{\mathrm{T}}\mathbf{V}\mathbf{x} + 2\mathbf{u}^{\mathrm{T}}\mathbf{x} + f \tag{0.7}$$

By comparison we obtain:

$$\mathbf{V} = \begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix} \tag{0.8}$$

$$\mathbf{u} = \begin{pmatrix} -\frac{1}{2} \\ -\frac{1}{2} \end{pmatrix} \tag{0.9}$$

$$f = 0 \tag{0.10}$$

We eigen decompose $\mathbf{V}$ (A.10) as

[*]Department of Electrical Engineering, Indian Institute of Technology, Hyderabad
[†]Department of Artifical Intelligence, Indian Institute of Technology, Hyderabad
[‡]Department of Electrical Engineering, Indian Institute of Technology, Hyderabad

$$\mathbf{V} = \mathbf{P}\mathbf{D}\mathbf{P}^{\mathrm{T}} \tag{0.11}$$

$$\mathbf{P} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \tag{0.12}$$

$$\mathbf{D} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{-1}{2} \end{pmatrix} \tag{0.13}$$

And convert the conic into a standard conic (A.50) to make it simpler to solve using affine transformations (A.3).

$$\mathbf{y}^{\mathrm{T}} \left(\frac{\mathbf{D}}{f_0} \right) \mathbf{y} = 1 \tag{0.14}$$

$$\mathbf{x} = \mathbf{P}\mathbf{y} + \mathbf{c} \tag{0.15}$$

Where

$$f_0 = \mathbf{u}^{\mathrm{T}}\mathbf{V}^{-1}\mathbf{u} - f = 1 \tag{0.16}$$

$$\mathbf{c} = -\mathbf{V}^{-1}\mathbf{u} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{0.17}$$

Since eigenvalues of $\mathbf{D}$ are $\lambda_1 = \frac{1}{2}$ and $\lambda_2 = -\frac{1}{2}$ We use another transformation to shift the negative eigenvalue of $\mathbf{D}$ to get the hyperbola in standard form, finally giving us

$$\mathbf{z}^{\mathrm{T}} \left(\frac{\mathbf{D_0}}{f_0} \right) \mathbf{z} = 1 \tag{0.18}$$

$$j(\mathbf{z}) = \mathbf{z}^{\mathrm{T}}\mathbf{D_0}\mathbf{z} - f_0 = 0 \tag{0.19}$$

$$\mathbf{y} = \mathbf{P}_0\mathbf{z} \tag{0.20}$$

Here $\mathbf{P}_0$ is the reflection matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $\mathbf{D}_0$ is defined as

$$\mathbf{D} = \mathbf{P}_0\mathbf{D}_0\mathbf{P}_0^{\mathrm{T}} \tag{0.21}$$

$$\mathbf{P}_0^{\mathrm{T}}\mathbf{D}\mathbf{P}_0 = \mathbf{D}_0 \tag{0.22}$$

$$\mathbf{D}_0 = \begin{pmatrix} \frac{-1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \tag{0.23}$$

Now we solve to find the endpoints of the latus recta of this standard conic (Appendix A.3.3)

$$\mathbf{n} = \sqrt{\lambda_2}\mathbf{p}_1 = \begin{pmatrix} 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix} \tag{0.24}$$

$$e = \sqrt{1 - \frac{\lambda_1}{\lambda_2}} = \sqrt{2} \tag{0.25}$$

$$c = \frac{e\mathbf{u}^{\mathrm{T}}\mathbf{n} \pm \sqrt{e^2(\mathbf{u}^T\mathbf{n})^2 - \lambda_2(e^2-1)(\|\mathbf{u}\|^2 - \lambda_2 f)}}{\lambda_2 e(e^2-1)} \tag{0.26}$$

$$= \pm\frac{1}{\sqrt{2}} \tag{0.27}$$

$$\mathbf{F} = \frac{ce^2\mathbf{n} - \mathbf{u}}{\lambda_2} = \pm 2\mathbf{e}_2 \tag{0.28}$$

$$\tag{0.29}$$

Equation of latus recta is

$$\mathbf{n}^{\mathrm{T}}\mathbf{x} = \mathbf{n}^{\mathrm{T}}\mathbf{F} \tag{0.30}$$
$$\equiv \mathbf{n}^{\mathrm{T}}\mathbf{x} = \sqrt{2} \tag{0.31}$$
$$\equiv \mathbf{x} = \mathbf{h} + k\mathbf{m} \tag{0.32}$$
$$\mathbf{h} = \begin{pmatrix} 0 \\ \pm 2 \end{pmatrix} \tag{0.33}$$
$$\mathbf{m} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{0.34}$$

Let $\hat{\mathbf{z}}$ be the endpoints of the latus recta i.e the intersection of (0.32) with the conic (0.19), solved in A.3.8

$$k = \frac{1}{\mathbf{m}^{\mathrm{T}}\mathbf{D_0}\mathbf{m}}\left(-\mathbf{m}^{\mathrm{T}}(\mathbf{D_0}\mathbf{h} + \mathbf{u_0}) \pm \sqrt{[\mathbf{m}^{\mathrm{T}}(\mathbf{D_0}\mathbf{h} + \mathbf{u_0})]^2 - j(\mathbf{h})(\mathbf{m}^{\mathrm{T}}\mathbf{D_0}\mathbf{m})}\right) \tag{0.35}$$
$$= \pm\sqrt{2} \tag{0.36}$$
$$\therefore \hat{\mathbf{z}} = \begin{pmatrix} \pm\sqrt{2} \\ \pm 2 \end{pmatrix} \tag{0.37}$$

Now transforming $\hat{\mathbf{z}}$ back using (0.15), (0.20) to get endpoints of original conic $\hat{\mathbf{x}}$.

$$\hat{\mathbf{x}} = \mathbf{P}\left(\mathbf{P}_0\hat{\mathbf{z}}\right) + \mathbf{c} \tag{0.38}$$

Which gives us values of $\hat{\mathbf{x}}$ to be

$$\hat{\mathbf{x}}_1 = \begin{pmatrix} 2+\sqrt{2} \\ \sqrt{2} \end{pmatrix}$$
$$\hat{\mathbf{x}}_2 = \begin{pmatrix} \sqrt{2} \\ 2+\sqrt{2} \end{pmatrix}$$
$$\hat{\mathbf{x}}_3 = \begin{pmatrix} 2-\sqrt{2} \\ -\sqrt{2} \end{pmatrix}$$
$$\hat{\mathbf{x}}_4 = \begin{pmatrix} -\sqrt{2} \\ 2-\sqrt{2} \end{pmatrix}$$

We can plot out the original conic in (0.7) to verify our solution

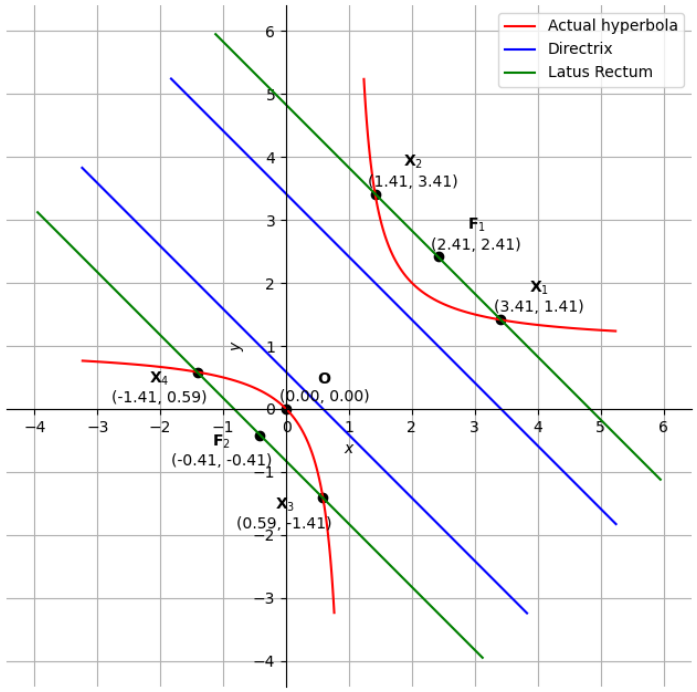

Figure 0.1

Out of these, $\hat{\mathbf{x}}_3$ and $\hat{\mathbf{x}}_4$ are unfit to be used as values for (a,b) as negative values of a or b will cause $f(t)$ to not be finite and thus the integral in (0.1c) will not converge and condition will not be met.

Lastly, We plot the function $f(t)$ for $(a,b) = \hat{\mathbf{x}}_1$ and $(a,b) = \hat{\mathbf{x}}_2$

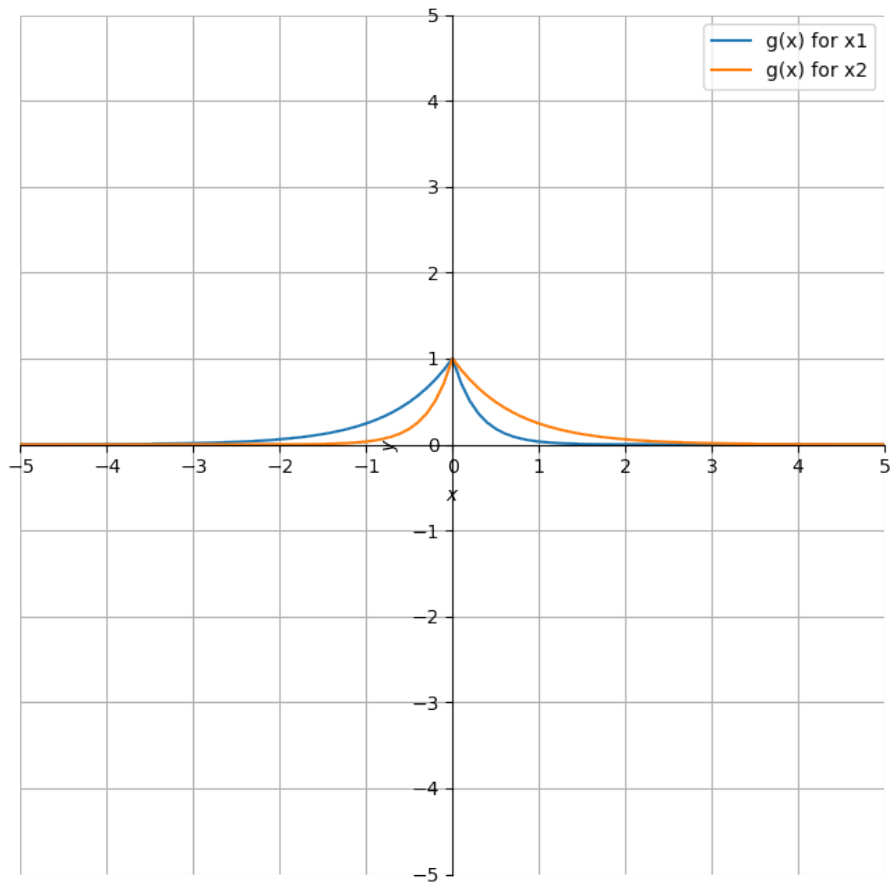

Figure 0.2

1 Derivation of Truncation Points

To plot the function $f(t)$ effectively, we define truncation points t_1 and t_2 such that a fraction p of the total area is retained. Given $f(t)$ from (0.1a) and (0.1b):

$$f(t) = \begin{cases} e^{bt}, & t < 0 \\ e^{-at}, & t > 0 \end{cases} \tag{1.1}$$

The total area is $A = \int_{-\infty}^{\infty} f(t)dt = \frac{1}{a} + \frac{1}{b} = 1$. To retain p area, we seek t_1 and t_2 such that the area in each excluded tail is $\frac{1-p}{2}$.

1.1 Lower Bound

Setting the area of the left tail to the excluded portion:

$$\int_{-\infty}^{t_1} e^{bt}\, dt = \frac{1-p}{2} \tag{1.2}$$

$$\left[\frac{e^{bt}}{b}\right]_{-\infty}^{t_1} = \frac{1-p}{2} \tag{1.3}$$

$$\frac{e^{bt_1}}{b} = \frac{1-p}{2} \tag{1.4}$$

$$t_1 = \frac{1}{b}\ln\left(\frac{b(1-p)}{2}\right) \tag{1.5}$$

1.2 Upper Bound

Setting the area of the right tail to the excluded portion:

$$\int_{t_2}^{\infty} e^{-at}\,dt = \frac{1-p}{2} \tag{1.6}$$

$$\left[-\frac{e^{-at}}{a}\right]_{t_2}^{\infty} = \frac{1-p}{2} \tag{1.7}$$

$$\frac{e^{-at_2}}{a} = \frac{1-p}{2} \tag{1.8}$$

$$t_2 = -\frac{1}{a}\ln\left(\frac{a(1-p)}{2}\right) \tag{1.9}$$

1.3 Truncation Percentage

A standard way to calculate the appropriate value of p is to express it in terms of the amplitude threshold ε. This means that the graph is truncated when the signal's amplitude drops to a small fraction, ε, of its peak value.

Let the truncation points t_1 and t_2 be defined as the exact points where the function decays to ε:

$$f(t_1) = \varepsilon \quad \text{and} \quad f(t_2) = \varepsilon$$

For the left side ($t < 0$):

$$e^{bt_1} = \varepsilon \implies t_1 = \frac{\ln(\varepsilon)}{b}$$

For the right side ($t > 0$):

$$e^{-at_2} = \varepsilon \implies -at_2 = \ln(\varepsilon) \implies t_2 = -\frac{\ln(\varepsilon)}{a}$$

$$A_{\text{left_excluded}} = \int_{-\infty}^{t_1} e^{bt}\,dt = \left[\frac{e^{bt}}{b}\right]_{-\infty}^{t_1} = \frac{e^{bt_1}}{b} = \frac{\varepsilon}{b}$$

$$A_{\text{right_excluded}} = \int_{t_2}^{\infty} e^{-at}\,dt = \left[-\frac{e^{-at}}{a}\right]_{t_2}^{\infty} = \frac{e^{-at_2}}{a} = \frac{\varepsilon}{a}$$

Thus we have:

$$A_{\text{excluded}} = \int_{-\infty}^{t_1} e^{bt}\,dt + \int_{t_2}^{\infty} e^{-at}\,dt = \frac{\varepsilon}{b} + \frac{\varepsilon}{a} = \varepsilon\left(\frac{1}{a} + \frac{1}{b}\right)$$

From (0.5):

$$A_{\text{excluded}} = \varepsilon$$

Since the retained area p is $1 - A_{\text{excluded}}$:

$$p = 1 - \varepsilon \tag{1.10}$$

1.4 Choice of Threshold

A reasonable choice for the threshold condition for the bounds t_1 and t_2 is the point where the signal undergoes a -30dB amplitude attenuation

Amplitude in decibels is calculated as $20\log_{10}(\varepsilon)$

$$20\log_{10}(\varepsilon) = -30 \tag{1.11}$$

$$\varepsilon = 10^{-1.5} \approx 0.03162 \tag{1.12}$$

From (1.10)

$$p = 1 - 0.03162 = \mathbf{0.96838}$$

Thus we have

$$t_1 = \frac{1}{b}\ln\left(\frac{b(1-p)}{2}\right) = \frac{1}{\sqrt{2}}\ln\left(\frac{\sqrt{2}(1-0.96838)}{2}\right) \approx -2.6874 \tag{1.13}$$

$$t_2 = -\frac{1}{a}\ln\left(\frac{a(1-p)}{2}\right) = -\frac{1}{2+\sqrt{2}}\ln\left(\frac{(2+\sqrt{2})(1-0.96838)}{2}\right) \approx 0.8550 \tag{1.14}$$

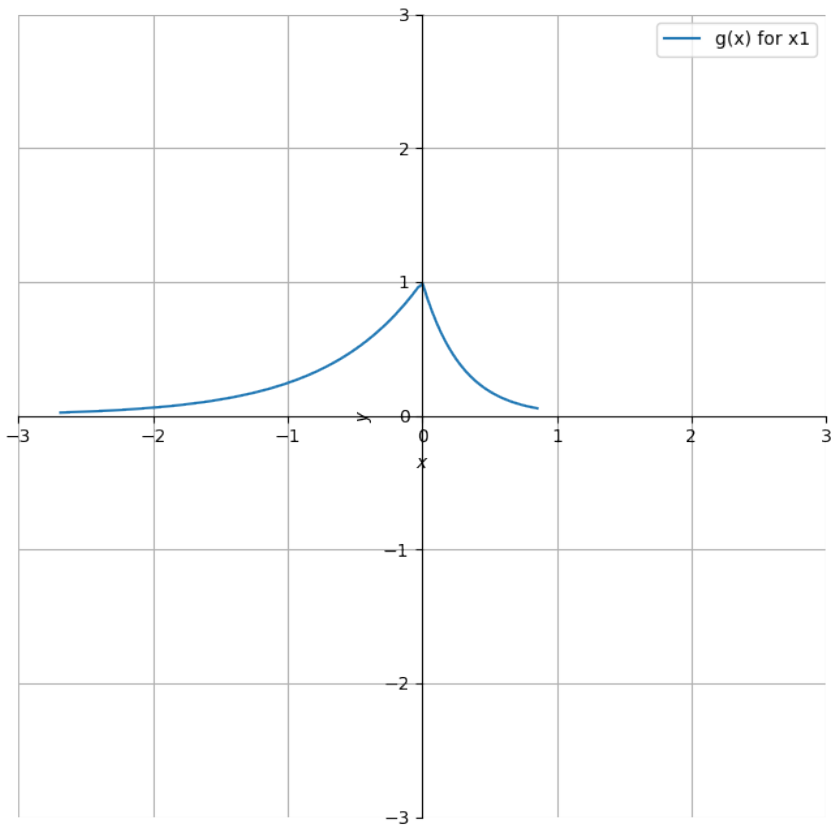

Figure 1.1

A Derivations

A.1 Linear Forms

A.1.1. Let $\mathbf{Q}$ be the foot of the perpendicular from $\mathbf{P}$ to the line

$$\mathbf{n}^\top \mathbf{x} = c \tag{A.1}$$

Then

$$PQ = \frac{\left|\mathbf{n}^\top \mathbf{P} - c\right|}{\|\mathbf{n}\|} \tag{A.2}$$

A.1.2. The affine transformation is given by

$$\mathbf{x} = \mathbf{P}\mathbf{y} + \mathbf{c} \tag{A.3}$$

where $\mathbf{c}$ is the translation vector.

A.2 Eigenvalues and Eigenvectors

A.2.1. The roots of the equation are known as the eigenvalues of $\mathbf{\Sigma}$.

$$\left|\lambda \mathbf{I} - \mathbf{\Sigma}\right| = 0 \tag{A.4}$$

A.2.2. The values of $\mathbf{p}$ that satisfy

$$\mathbf{\Sigma}\mathbf{p} = \lambda \mathbf{p} \tag{A.5}$$

are known as the eigenvectors of Σ.

A.2.3. Define

$$\mathbf{D} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}, \tag{A.6}$$

$$\mathbf{P} = \begin{pmatrix} \frac{\mathbf{p}_1}{\|\mathbf{p}_1\|} & \frac{\mathbf{p}_2}{\|\mathbf{p}_2\|} \end{pmatrix} \tag{A.7}$$

A.2.4. Note that

$$\mathbf{P}^\top = \mathbf{P}^{-1}. \tag{A.8}$$

$\mathbf{P}$ is defined to be an orthogonal matrix.

A.2.5. Note that

$$\mathbf{P}^\top \mathbf{\Sigma} \mathbf{P} = \mathbf{D}, \tag{A.9}$$

This is known as the spectral (eigenvalue) decomposition of a symmetric matrix

A.2.6. For a symmetric matrix, from (A.9), we have the eigendecomposition

$$\mathbf{V} = \mathbf{P}\mathbf{D}\mathbf{P}^\top \tag{A.10}$$

where

$$\mathbf{P} = \begin{pmatrix} \mathbf{p}_1 & \mathbf{p}_2 \end{pmatrix}, \quad \mathbf{P}^\top \mathbf{P} = \mathbf{I} \tag{A.11}$$

$$\mathbf{D} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \tag{A.12}$$

A.3 Conic Section

A.3.1. Let $\mathbf{q}$ be a point such that the ratio of its distance from a fixed point $\mathbf{F}$ and the distance (d) from a fixed line

$$L : \mathbf{n}^\top \mathbf{x} = c \tag{A.13}$$

is constant, given by

$$\frac{\|\mathbf{q}-\mathbf{F}\|}{d} = e \tag{A.14}$$

The locus of $\mathbf{q}$ is known as a conic section. The line L is known as the directrix and the point $\mathbf{F}$ is the focus. e is defined to be the eccentricity of the conic.

(a) For $e = 1$, the conic is a parabola
(b) For $e < 1$, the conic is an ellipse
(c) For $e > 1$, the conic is a hyperbola

A.3.2. The equation of a conic with directrix $\mathbf{n}^\top \mathbf{x} = c$, eccentricity e and focus $\mathbf{F}$ is given by

$$\mathrm{g}(\mathbf{x}) = \mathbf{x}^\top \mathbf{V}\mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \tag{A.15}$$

where

$$\mathbf{V} = \|\mathbf{n}\|^2 \mathbf{I} - e^2 \mathbf{n}\mathbf{n}^\top, \tag{A.16}$$
$$\mathbf{u} = ce^2 \mathbf{n} - \|\mathbf{n}\|^2 \mathbf{F}, \tag{A.17}$$
$$f = \|\mathbf{n}\|^2 \|\mathbf{F}\|^2 - c^2 e^2 \tag{A.18}$$

Solution: Using Definition A.3.1 and (A.2), for any point $\mathbf{x}$ on the conic,

$$\|\mathbf{x}-\mathbf{F}\|^2 = e^2 \frac{\left(\mathbf{n}^\top \mathbf{x} - c\right)^2}{\|\mathbf{n}\|^2} \tag{A.19}$$
$$\implies \|\mathbf{n}\|^2 (\mathbf{x}-\mathbf{F})^\top (\mathbf{x}-\mathbf{F}) = e^2 \left(\mathbf{n}^\top \mathbf{x} - c\right)^2 \tag{A.20}$$
$$\implies \|\mathbf{n}\|^2 \left(\mathbf{x}^\top \mathbf{x} - 2\mathbf{F}^\top \mathbf{x} + \|\mathbf{F}\|^2\right) = e^2 \left(c^2 + \left(\mathbf{n}^\top \mathbf{x}\right)^2 - 2c\mathbf{n}^\top \mathbf{x}\right) \tag{A.21}$$
$$= e^2 \left(c^2 + \left(\mathbf{x}^\top \mathbf{n}\mathbf{n}^\top \mathbf{x}\right) - 2c\mathbf{n}^\top \mathbf{x}\right) \tag{A.22}$$

which can be expressed as (A.15) after simplification.

A.3.3. The eccentricity, directrices and foci of (A.15) are given by

$$e = \sqrt{1 - \frac{\lambda_1}{\lambda_2}} \tag{A.23}$$
$$\mathbf{n} = \sqrt{\lambda_2}\mathbf{p}_1,$$
$$c = \begin{cases} \frac{e\mathbf{u}^\top \mathbf{n} \pm \sqrt{e^2\left(\mathbf{u}^\top \mathbf{n}\right)^2 - \lambda_2\left(e^2-1\right)\left(\|\mathbf{u}\|^2 - \lambda_2 f\right)}}{\lambda_2 e\left(e^2-1\right)} & e \neq 1 \\ \frac{\|\mathbf{u}\|^2 - \lambda_2 f}{2\mathbf{u}^\top \mathbf{n}} & e = 1 \end{cases} \tag{A.24}$$
$$\mathbf{F} = \frac{ce^2 \mathbf{n} - \mathbf{u}}{\lambda_2} \tag{A.25}$$

Solution: From (A.16), using the fact that $\mathbf{V}$ is symmetric with $\mathbf{V} = \mathbf{V}^\top$,

$$\mathbf{V}^\top \mathbf{V} = \left(\|\mathbf{n}\|^2 \mathbf{I} - e^2 \mathbf{n}\mathbf{n}^\top\right)^\top \left(\|\mathbf{n}\|^2 \mathbf{I} - e^2 \mathbf{n}\mathbf{n}^\top\right) \tag{A.26}$$

$$\implies \mathbf{V}^2 = \|\mathbf{n}\|^4 \mathbf{I} + e^4 \mathbf{n}\mathbf{n}^\top \mathbf{n}\mathbf{n}^\top - 2e^2 \|\mathbf{n}\|^2 \mathbf{n}\mathbf{n}^\top \tag{A.27}$$

$$= \|\mathbf{n}\|^4 \mathbf{I} + e^4 \|\mathbf{n}\|^2 \mathbf{n}\mathbf{n}^\top - 2e^2 \|\mathbf{n}\|^2 \mathbf{n}\mathbf{n}^\top \tag{A.28}$$

$$= \|\mathbf{n}\|^4 \mathbf{I} + e^2 \left(e^2 - 2\right) \|\mathbf{n}\|^2 \mathbf{n}\mathbf{n}^\top \tag{A.29}$$

$$= \|\mathbf{n}\|^4 \mathbf{I} + \left(e^2 - 2\right) \|\mathbf{n}\|^2 \left(\|\mathbf{n}\|^2 \mathbf{I} - \mathbf{V}\right) \tag{A.30}$$

which can be expressed as

$$\mathbf{V}^2 + \left(e^2 - 2\right) \|\mathbf{n}\|^2 \mathbf{V} - \left(e^2 - 1\right) \|\mathbf{n}\|^4 \mathbf{I} = 0 \tag{A.31}$$

Using the Cayley-Hamilton theorem, (A.31) results in the characteristic equation,

$$\lambda^2 - \left(2 - e^2\right) \|\mathbf{n}\|^2 \lambda + \left(1 - e^2\right) \|\mathbf{n}\|^4 = 0 \tag{A.32}$$

which can be expressed as

$$\left(\frac{\lambda}{\|\mathbf{n}\|^2}\right)^2 - \left(2 - e^2\right) \left(\frac{\lambda}{\|\mathbf{n}\|^2}\right) + \left(1 - e^2\right) = 0 \tag{A.33}$$

$$\implies \frac{\lambda}{\|\mathbf{n}\|^2} = 1 - e^2, 1 \tag{A.34}$$

$$\text{or, } \lambda_2 = \|\mathbf{n}\|^2, \lambda_1 = \left(1 - e^2\right) \lambda_2 \tag{A.35}$$

From (A.35), the eccentricity of (A.15) is given by (A.23). Multiplying both sides of (A.16) by $\mathbf{n}$,

$$\mathbf{V}\mathbf{n} = \|\mathbf{n}\|^2 \mathbf{n} - e^2 \mathbf{n}\mathbf{n}^\top \mathbf{n} \tag{A.36}$$

$$= \|\mathbf{n}\|^2 \left(1 - e^2\right) \mathbf{n} \tag{A.37}$$

$$= \lambda_1 \mathbf{n} \tag{A.38}$$

$$\tag{A.39}$$

from (A.35). Thus, λ_1 is the corresponding eigenvalue for $\mathbf{n}$. From (A.7) and (A.39), this implies that

$$\mathbf{p}_1 = \frac{\mathbf{n}}{\|\mathbf{n}\|} \tag{A.40}$$

$$\text{or, } \mathbf{n} = \|\mathbf{n}\| \mathbf{p}_1 = \sqrt{\lambda_2} \mathbf{p}_1 \tag{A.41}$$

from (A.35) . From (A.17) and (A.35),

$$\mathbf{F} = \frac{ce^2 \mathbf{n} - \mathbf{u}}{\lambda_2} \tag{A.42}$$

$$\implies \|\mathbf{F}\|^2 = \frac{\left(ce^2 \mathbf{n} - \mathbf{u}\right)^\top \left(ce^2 \mathbf{n} - \mathbf{u}\right)}{\lambda_2^2} \tag{A.43}$$

$$\implies \lambda_2^2 \|\mathbf{F}\|^2 = c^2 e^4 \lambda_2 - 2ce^2 \mathbf{u}^\top \mathbf{n} + \|\mathbf{u}\|^2 \tag{A.44}$$

Also, (A.18) can be expressed as

$$\lambda_2 \|\mathbf{F}\|^2 = f + c^2 e^2 \tag{A.45}$$

From (A.44) and (A.45),

$$c^2 e^4 \lambda_2 - 2ce^2 \mathbf{u}^\top \mathbf{n} + \|\mathbf{u}\|^2 = \lambda_2 \left(f + c^2 e^2\right) \tag{A.46}$$

$$\implies \lambda_2 e^2 \left(e^2 - 1\right) c^2 - 2ce^2 \mathbf{u}^\top \mathbf{n} + \|\mathbf{u}\|^2 - \lambda_2 f = 0 \tag{A.47}$$

yielding (A.25).

A.3.4. (A.15) represents

(a) a parabola for $\left|\mathbf{V}\right| = 0$,
(b) ellipse for $\left|\mathbf{V}\right| > 0$ and
(c) hyperbola for $\left|\mathbf{V}\right| < 0$.

Solution: From (A.23),

$$\frac{\lambda_1}{\lambda_2} = 1 - e^2 \tag{A.48}$$

Also,

$$\left|\mathbf{V}\right| = \lambda_1 \lambda_2 \tag{A.49}$$

yielding Table A.3.4.

Eccentricity	**Conic**	**Eigenvalue**	**Determinant**
$e = 1$	Parabola	$\lambda_1 = 0$	$\left\|\mathbf{V}\right\| = 0$
$e < 1$	Ellipse	$\lambda_1 > 0, \lambda_2 > 0$	$\left\|\mathbf{V}\right\| > 0$
$e > 1$	Hyperbola	$\lambda_1 < 0, \lambda_2 > 0$	$\left\|\mathbf{V}\right\| < 0$

Table A.3.4

A.3.5. Using the affine transformation in (A.3), the conic in (A.15) can be expressed in standard form ;as

$$\mathbf{y}^\top \left(\frac{\mathbf{D}}{f_0}\right) \mathbf{y} = 1 \qquad |\mathbf{V}| \neq 0 \tag{A.50}$$

$$\mathbf{y}^\top \mathbf{D} \mathbf{y} = -\eta \mathbf{e}_1^\top \mathbf{y} \qquad |\mathbf{V}| = 0 \tag{A.51}$$

where

$$f_0 = \mathbf{u}^\top \mathbf{V}^{-1} \mathbf{u} - f \neq 0 \tag{A.52}$$

$$\eta = 2\mathbf{u}^\top \mathbf{p}_1 \tag{A.53}$$

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{A.54}$$

Solution: Using (A.3), (A.15) can be expressed as

$$(\mathbf{P}\mathbf{y} + \mathbf{c})^\top \mathbf{V} (\mathbf{P}\mathbf{y} + \mathbf{c}) + 2\mathbf{u}^\top (\mathbf{P}\mathbf{y} + \mathbf{c}) + f = 0, \tag{A.55}$$

yielding

$$\mathbf{y}^\top \mathbf{P}^\top \mathbf{V} \mathbf{P} \mathbf{y} + 2 (\mathbf{V}\mathbf{c} + \mathbf{u})^\top \mathbf{P}\mathbf{y} + \mathbf{c}^\top \mathbf{V}\mathbf{c} + 2\mathbf{u}^\top \mathbf{c} + f = 0 \tag{A.56}$$

From (A.56) and (A.9),

$$\mathbf{y}^\top \mathbf{D}\mathbf{y} + 2 (\mathbf{V}\mathbf{c} + \mathbf{u})^\top \mathbf{P}\mathbf{y} + \mathbf{c}^\top (\mathbf{V}\mathbf{c} + \mathbf{u}) + \mathbf{u}^\top \mathbf{c} + f = 0 \tag{A.57}$$

When $\mathbf{V}^{-1}$ exists, choosing

$$\mathbf{V}\mathbf{c} + \mathbf{u} = \mathbf{0}, \quad \text{or}, \mathbf{c} = -\mathbf{V}^{-1}\mathbf{u}, \tag{A.58}$$

and substituting (A.58) in (A.57) yields (A.50). When $|\mathbf{V}| = 0, \lambda_1 = 0$ and

$$\mathbf{V}\mathbf{p}_1 = 0, \mathbf{V}\mathbf{p}_2 = \lambda_2 \mathbf{p}_2. \tag{A.59}$$

Substituting (A.11) in (A.57),

$$\begin{aligned}
\mathbf{y}^\top \mathbf{D}\mathbf{y} + 2\left(\mathbf{c}^\top \mathbf{V} + \mathbf{u}^\top\right)\begin{pmatrix}\mathbf{p}_1 & \mathbf{p}_2\end{pmatrix}\mathbf{y} + \mathbf{c}^\top(\mathbf{V}\mathbf{c}+\mathbf{u}) + \mathbf{u}^\top\mathbf{c} + f = 0 \\
\implies \mathbf{y}^\top \mathbf{D}\mathbf{y} + 2\left(\left(\mathbf{c}^\top \mathbf{V} + \mathbf{u}^\top\right)\mathbf{p}_1 \left(\mathbf{c}^\top \mathbf{V} + \mathbf{u}^\top\right)\mathbf{p}_2\right)\mathbf{y} + \mathbf{c}^\top(\mathbf{V}\mathbf{c}+\mathbf{u}) + \mathbf{u}^\top\mathbf{c} + f = 0 \\
\implies \mathbf{y}^\top \mathbf{D}\mathbf{y} + 2\left(\mathbf{u}^\top\mathbf{p}_1 \quad \left(\lambda_2\mathbf{c}^\top + \mathbf{u}^\top\right)\mathbf{p}_2\right)\mathbf{y} + \mathbf{c}^\top(\mathbf{V}\mathbf{c}+\mathbf{u}) + \mathbf{u}^\top\mathbf{c} + f = 0
\end{aligned}$$

upon substituting from (A.59), yielding

$$\lambda_2 y_2^2 + 2\left(\mathbf{u}^\top\mathbf{p}_1\right)y_1 + 2y_2\left(\lambda_2\mathbf{c}+\mathbf{u}\right)^\top\mathbf{p}_2 + \mathbf{c}^\top(\mathbf{V}\mathbf{c}+\mathbf{u}) + \mathbf{u}^\top\mathbf{c} + f = 0 \tag{A.60}$$

which is the equation of a parabola. Thus, (A.60) can be expressed as (A.51) by choosing

$$\eta = 2\mathbf{u}^\top\mathbf{p}_1 \tag{A.61}$$

and $\mathbf{c}$ in (A.57) such that

$$2\mathbf{P}^\top(\mathbf{V}\mathbf{c}+\mathbf{u}) = \eta\begin{pmatrix}1\\0\end{pmatrix} \tag{A.62}$$

$$\mathbf{c}^\top(\mathbf{V}\mathbf{c}+\mathbf{u}) + \mathbf{u}^\top\mathbf{c} + f = 0 \tag{A.63}$$

A.3.6. The center/vertex of a conic section are given by

$$\mathbf{c} = -\mathbf{V}^{-1}\mathbf{u} \qquad |\mathbf{V}| \neq 0 \tag{A.64}$$

$$\begin{pmatrix}\mathbf{u}^\top + \frac{\eta}{2}\mathbf{p}_1^\top \\ \mathbf{V}\end{pmatrix}\mathbf{c} = \begin{pmatrix}-f \\ \frac{\eta}{2}\mathbf{p}_1 - \mathbf{u}\end{pmatrix} \qquad |\mathbf{V}| = 0 \tag{A.65}$$

Solution: $\because \mathbf{P}^\top\mathbf{P} = \mathbf{I}$, multiplying (A.62) by $\mathbf{P}$ yields

$$(\mathbf{V}\mathbf{c}+\mathbf{u}) = \frac{\eta}{2}\mathbf{p}_1, \tag{A.66}$$

which, upon substituting in (A.63) results in

$$\frac{\eta}{2}\mathbf{c}^\top\mathbf{p}_1 + \mathbf{u}^\top\mathbf{c} + f = 0 \tag{A.67}$$

(A.66) and (A.67) can be clubbed together to obtain (A.65).

A.3.7. In (A.3), substituting $\mathbf{y} = \mathbf{0}$, the center/vertex for the quadratic form is obtained as

$$\mathbf{x} = \mathbf{c}, \tag{A.68}$$

where $\mathbf{c}$ is derived as (A.64) and (A.65) in Appendix A.3.5.

A.3.8. The points of intersection of the line

$$L: \quad \mathbf{x} = \mathbf{h} + \kappa\mathbf{m} \quad \kappa \in \mathbb{R} \tag{A.69}$$

with the conic section in (A.15) are given by

$$\mathbf{x}_i = \mathbf{h} + \kappa_i\mathbf{m} \tag{A.70}$$

where

$$\kappa_i = \frac{1}{\mathbf{m}^\top\mathbf{V}\mathbf{m}}\left(-\mathbf{m}^\top(\mathbf{V}\mathbf{h}+\mathbf{u}) \pm \sqrt{\left[\mathbf{m}^\top(\mathbf{V}\mathbf{h}+\mathbf{u})\right]^2 - \mathrm{g}(\mathbf{h})\left(\mathbf{m}^\top\mathbf{V}\mathbf{m}\right)}\right) \tag{A.71}$$

Solution: Substituting (A.69) in (A.15),

$$(\mathbf{h}+\kappa\mathbf{m})^\top\mathbf{V}(\mathbf{h}+\kappa\mathbf{m}) + 2\mathbf{u}^\top(\mathbf{h}+\kappa\mathbf{m}) + f = 0 \tag{A.72}$$

$$\implies \kappa^2\mathbf{m}^\top\mathbf{V}\mathbf{m} + 2\kappa\mathbf{m}^\top(\mathbf{V}\mathbf{h}+\mathbf{u}) + \mathbf{h}^\top\mathbf{V}\mathbf{h} + 2\mathbf{u}^\top\mathbf{h} + f = 0 \tag{A.73}$$

$$\text{or, } \kappa^2\mathbf{m}^\top\mathbf{V}\mathbf{m} + 2\kappa\mathbf{m}^\top(\mathbf{V}\mathbf{h}+\mathbf{u}) + \mathrm{g}(\mathbf{h}) = 0 \tag{A.74}$$

for g defined in (A.15). Solving the above quadratic in (A.74) yields (A.71).